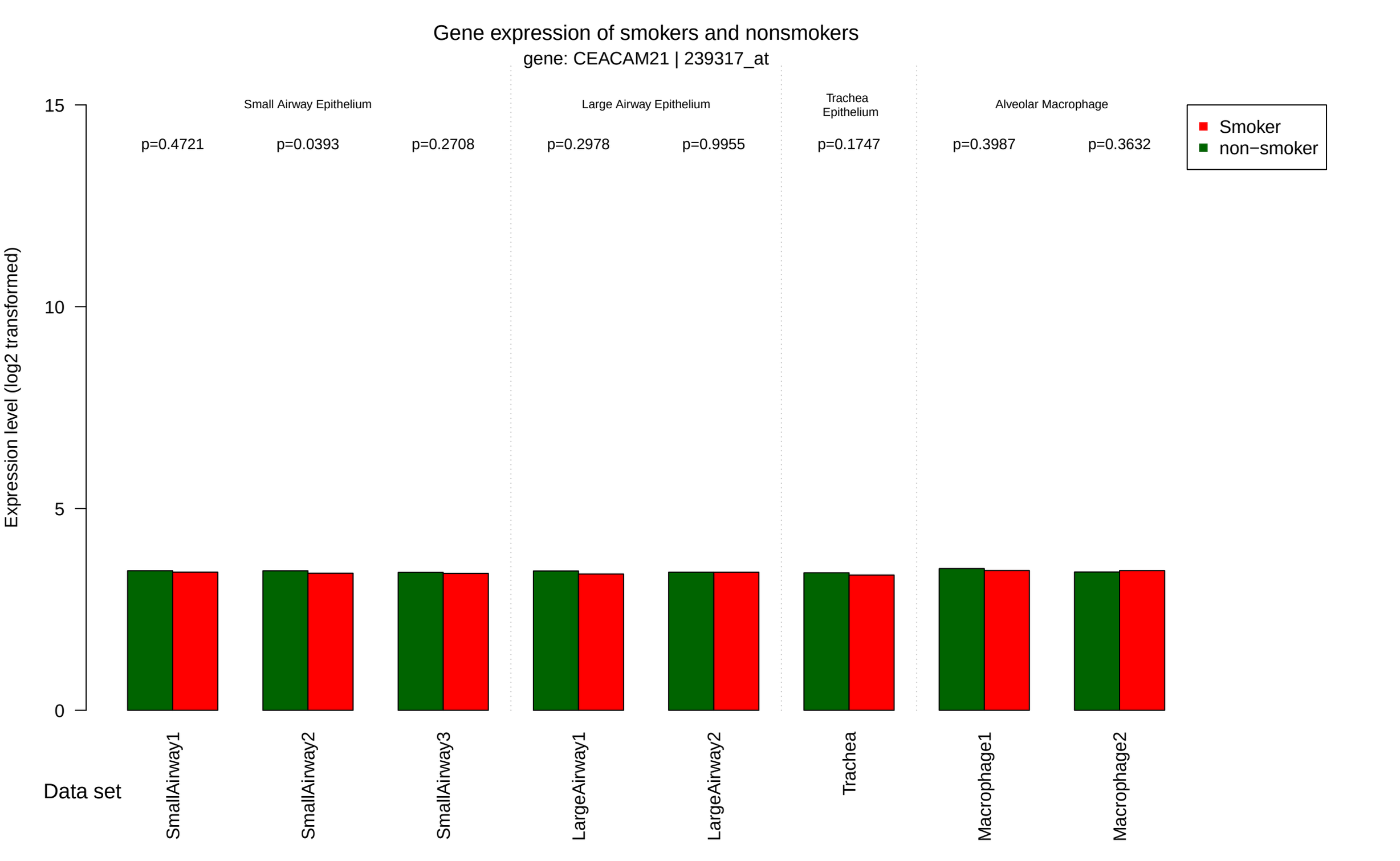
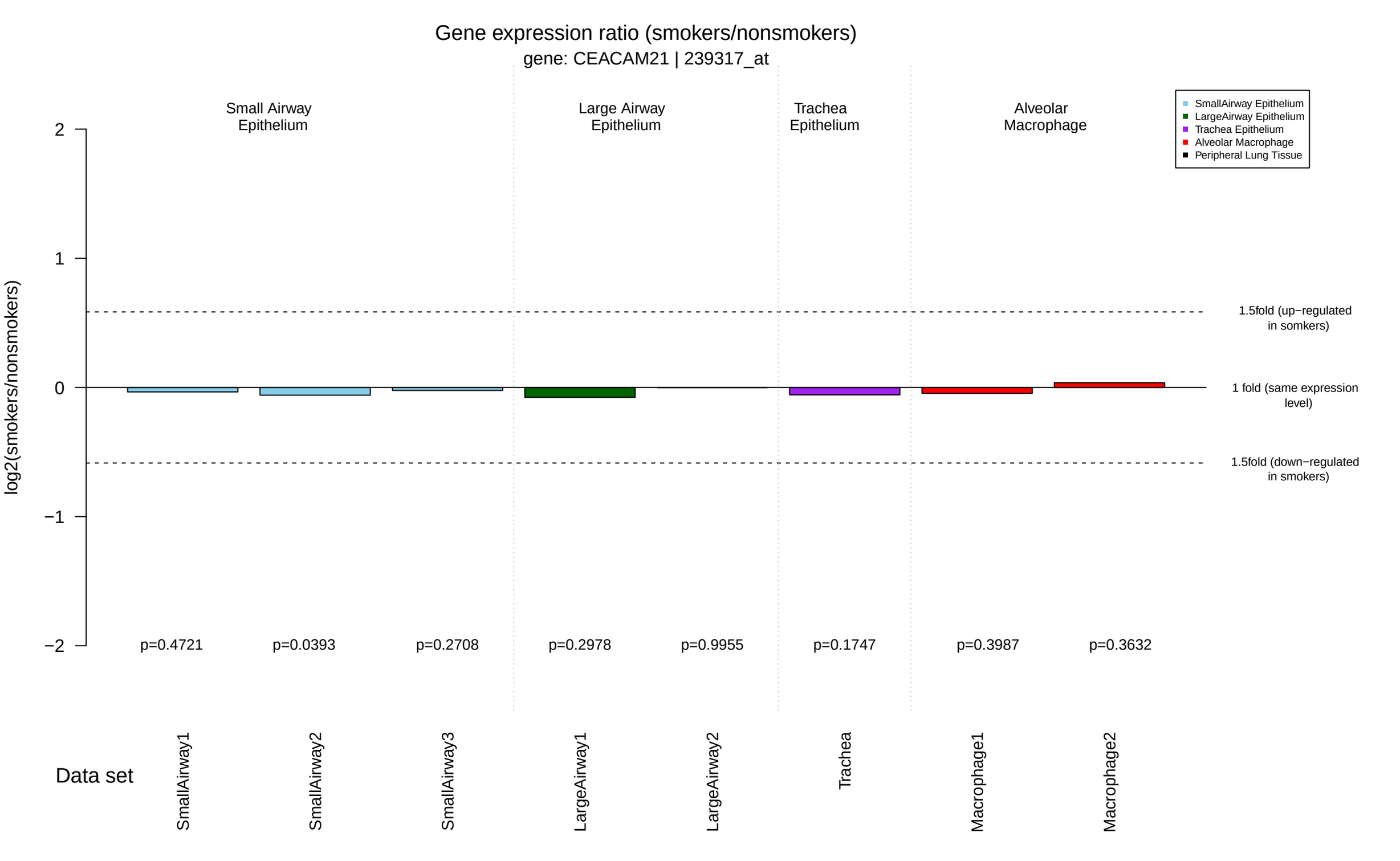
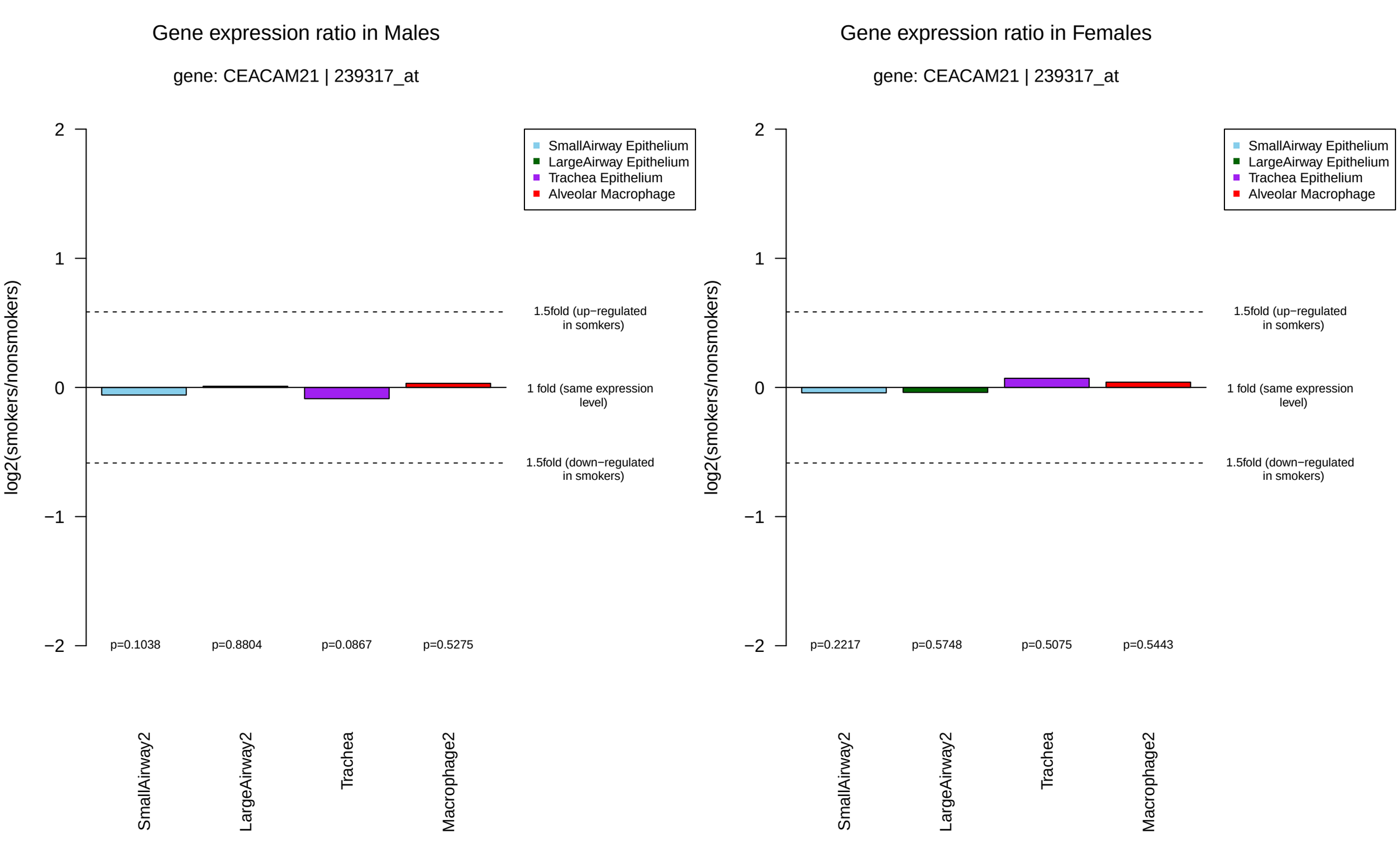
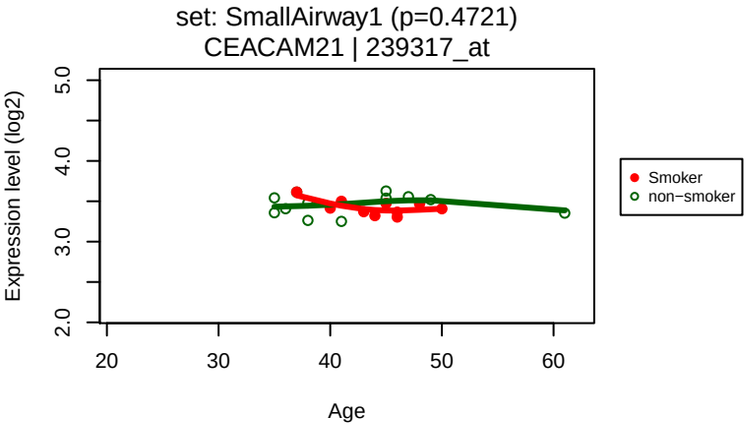
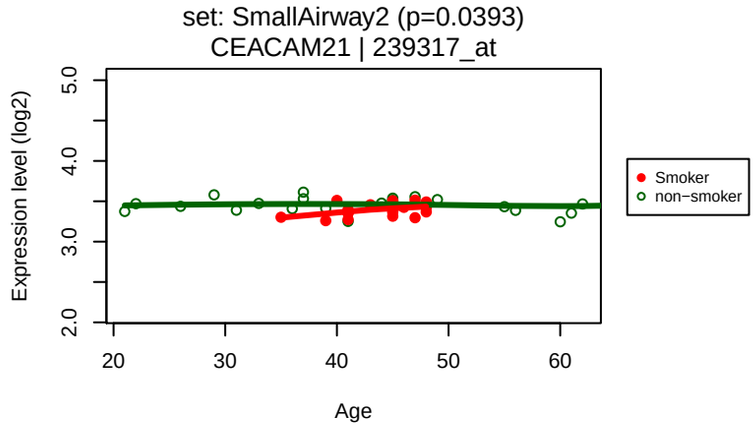
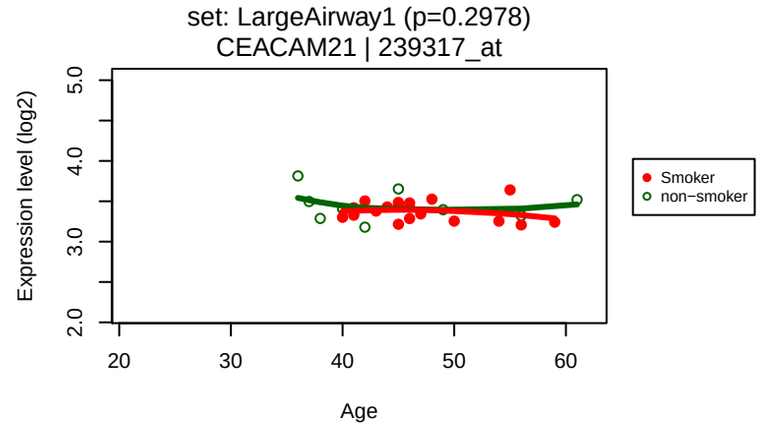

Gene expression of smokers and nonsmokers
gene: CEACAM21 | 239317_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4721
p=0.0393
p=0.2708
p=0.2978
p=0.9955
p=0.1747
p=0.3987
p=0.3632
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

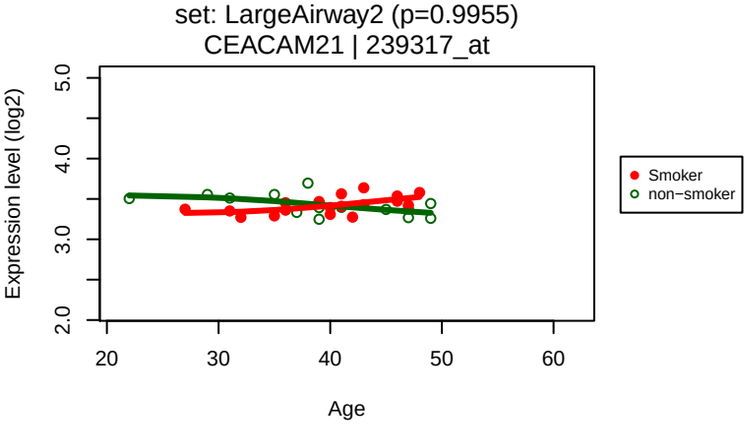

Gene expression ratio (smokers/nonsmokers)
gene: CEACAM21 | 239317_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4721
p=0.0393
p=0.2708
p=0.2978
p=0.9955
p=0.1747
p=0.3987
p=0.3632
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

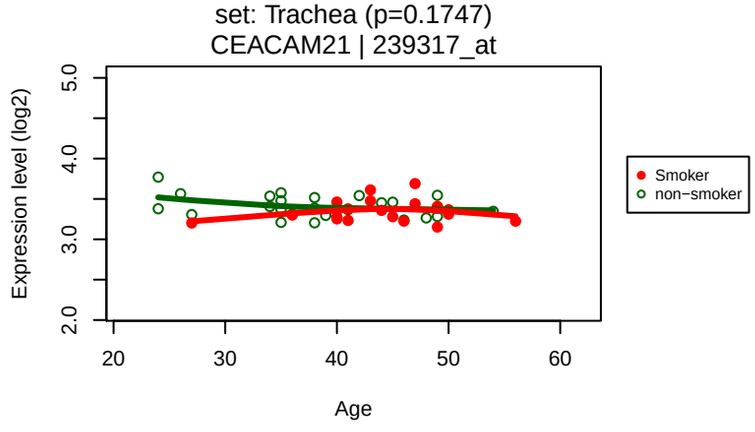

Gene expression ratio in Males
gene: CEACAM21 | 239317_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1038
p=0.8804
p=0.0867
p=0.5275
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CEACAM21 | 239317_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2217
p=0.5748
p=0.5075
p=0.5443
SmallAirway2
LargeAirway2
Trachea
Macrophage2

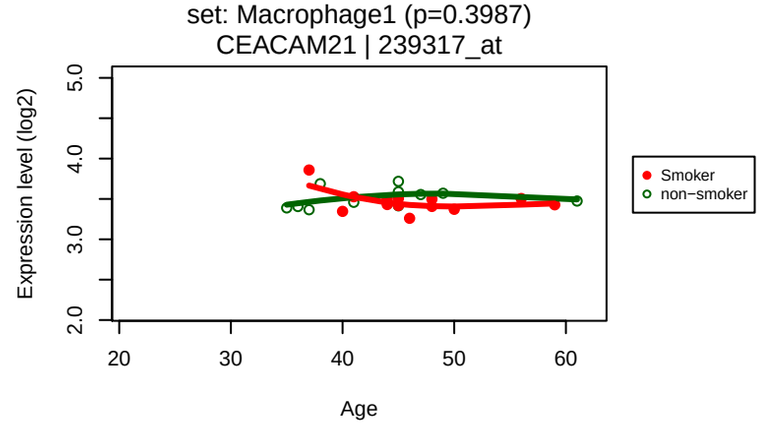

set: SmallAirway1 (p=0.4721)
CEACAM21 | 239317_at
Expression level (log2)
Age
Smoker
non−smoker

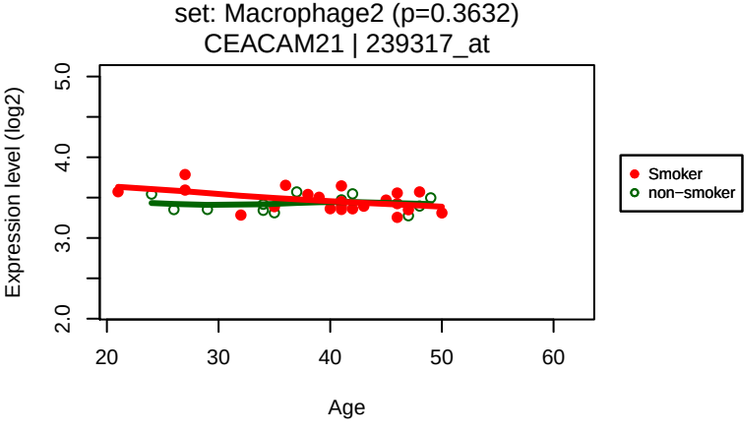

set: SmallAirway2 (p=0.0393)
CEACAM21 | 239317_at
Expression level (log2)
Age
Smoker
non−smoker

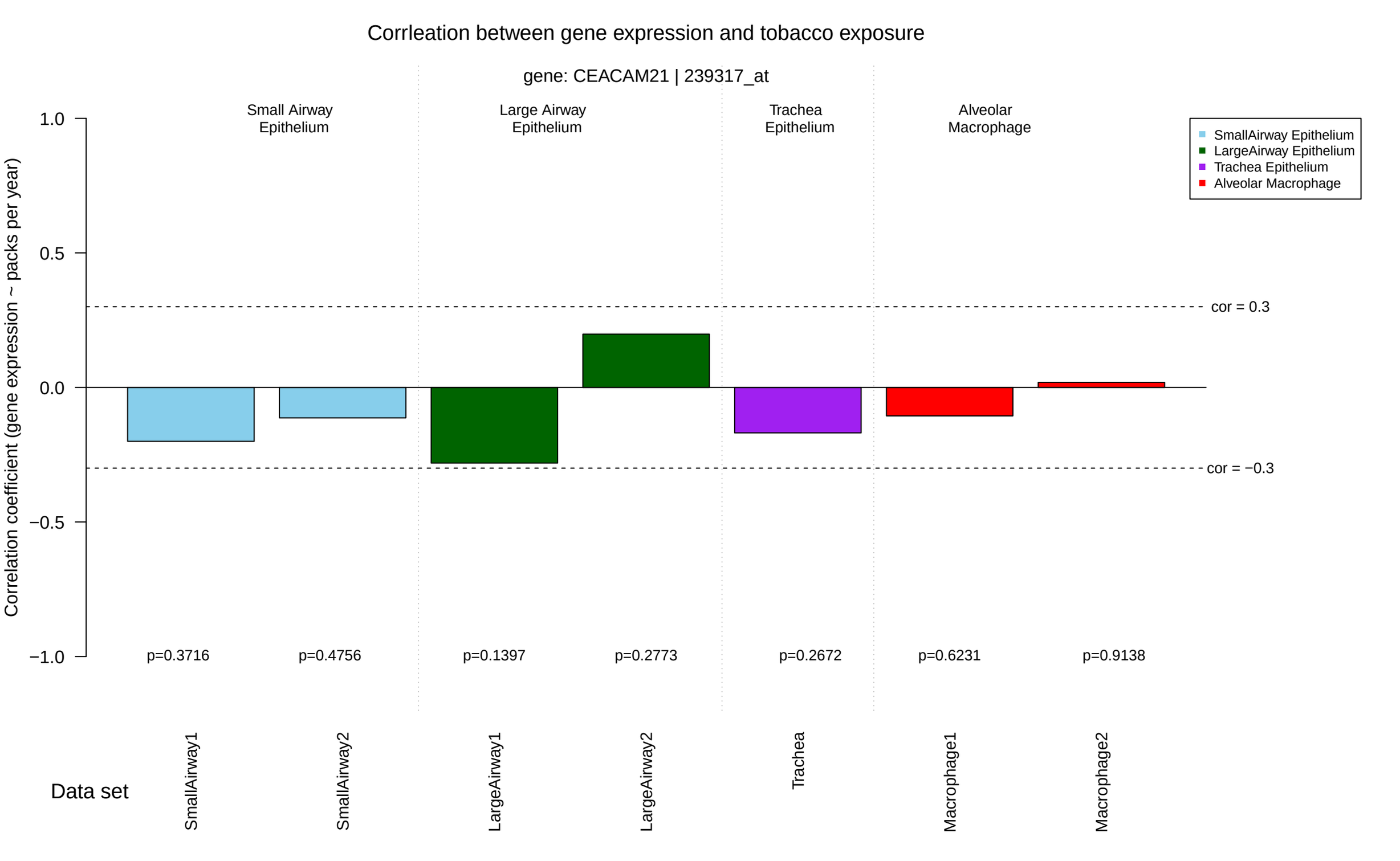

set: LargeAirway1 (p=0.2978)
CEACAM21 | 239317_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9955)
CEACAM21 | 239317_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1747)
CEACAM21 | 239317_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3987)
CEACAM21 | 239317_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3632)
CEACAM21 | 239317_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CEACAM21 | 239317_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3716
p=0.4756
p=0.1397
p=0.2773
p=0.2672
p=0.6231
p=0.9138
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2